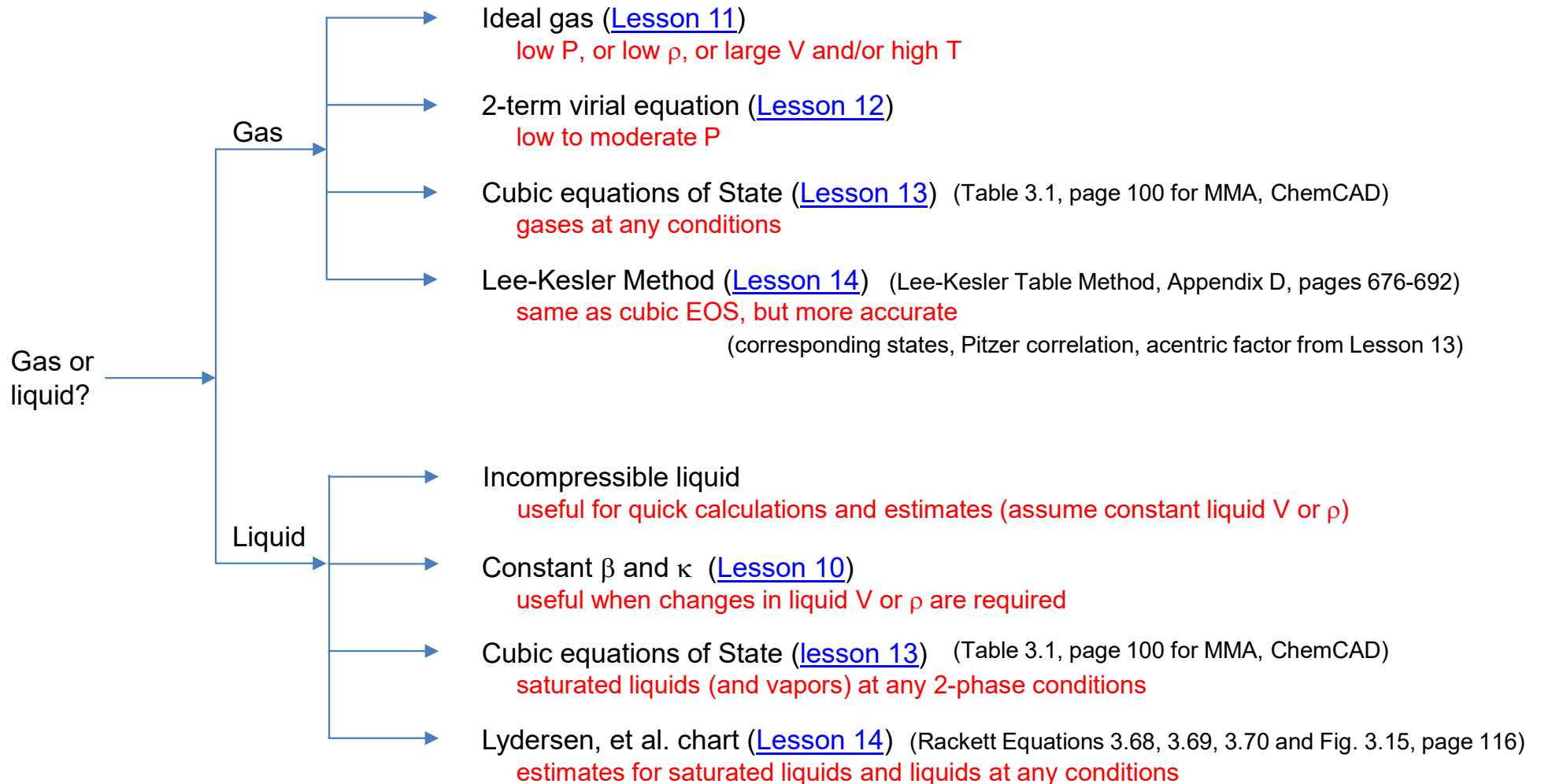


CH365 Chemical Engineering Thermodynamics

Lesson 15 Review

Equation of State Decision Tree

Be able to intelligently select an appropriate equation of state or generalized correlation for a given application.



IMPORTANT: Be able to apply cubic equations of state to calculation of work using $\int P dV$ or $\Delta(PV)$ given $P(V)$ from equation 3.41

Homework

2nd Virial Coefficients

Slide 4

Can be Derived from Theory

2-molecule pair-wise IMFs



Electrostatic forces
(H atom + H atom $\rightarrow$ H₂)

London dispersion forces
(He, Ne, Ar, Kr, Xe)

Problem 3.78

The *Boyle temperature* is the temperature for which:

$$\lim_{P \rightarrow 0} \left(\frac{\partial Z}{\partial P} \right)_T \equiv 0$$

- (a) Show that the second virial coefficient B is zero at the Boyle temperature.
- (b) Use the generalized correlation for B , Eqs. 3.58 to 3.62, to estimate the reduced Boyle temperature for simple fluids.